

Praveen Kumar S

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EDUCATION

Indian Institute of Information Technology Kottayam

Bachelor of Technology in Computer Science Engineering

2023 – 2027

CGPA: 8.69

Work Experience

AI Intern, Backspace Tech

December 2025 - May 2026

- Developed a document redaction pipeline using Tesseract and EasyOCR to automatically detect and mask sensitive customer banking information from invoice documents, ensuring data privacy compliance.
- Built an automated invoice processing system using PaddleOCR and Camelot for structured table extraction, integrated with Qwen LLM to intelligently parse and extract key fields from PDF invoices.
- Implemented an automated customer claims validation system using Qwen Vision-Language Model that analyzes product images alongside claim details to accept or reject customer claims, reducing manual review effort.
- Trained a YOLOv8 Medium object detection model to identify vehicle number plates and count axles in real-time at toll gates, enabling automated vehicle classification and FASTag-based toll collection.

Generative AI Intern, Granville Tech

July 2025 - October 2025

- Built a multi-agent RAG system to automate professional proposal generation. The system analyzes RFPs, conducts research, generates content, and learns from user feedback to continuously improve.
- Architected a multi-agent system featuring an RFP Analysis Agent, a Scoring Agent for quality control, a Guidance Agent for strategic insights, and a Mermaid Corrector Agent for automated syntax repair.
- Built a dual-memory system using a FAISS vector store for RAG-based retrieval and a networkx Knowledge Graph that learns from user feedback via a JSON log to improve performance over time.
- Implemented a self-correction loop where a dedicated Scoring Agent evaluates AI-generated content and triggers automatic rewrites until it meets quality standards, ensuring high-relevance output.
- Built and deployed a real-time, AI-powered recommendation system for a professional networking platform using GraphRAG, Neo4j, and Supabase to intelligently suggest user connections and drive engagement.

AI Intern, Infosys Springboard

September 2025 - October 2025

- Developed an end-to-end deep learning system to predict the Remaining Useful Life (RUL) of turbofan engines using the NASA CMAPSS time-series dataset.
- Engineered key features by calculating RUL targets for training data and implemented data cleaning techniques to remove constant-value sensors, improving data quality.
- Designed and trained a Long Short-Term Memory (LSTM) neural network using TensorFlow and Keras to model engine degradation patterns and predict failures.
- Evaluated model performance on an unseen test set, achieving a Root Mean Squared Error (RMSE) of approximately 30 cycles, providing a quantitative measure of prediction accuracy.

Research Intern, NIT Trichy

June 2025 - July 2025

- Performed audio and video-based sentiment analysis by converting speech signals into spectrogram images and extracting deep acoustic embeddings using TRILL and Wav2Vec, while leveraging ResNet and LSTM models to capture spatial-temporal features from video data for multimodal emotion understanding.
- Developed high-accuracy sentiment classification models on textual data, experimenting with multiple feature extraction, embedding techniques.
- Improved efficiency by using transformer models such as BERT and RoBERTa embedding with an LSTM architecture and achieved 99% accuracy outperforming traditional baselines.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, Java, HTML, CSS, PHP

Frameworks & Libraries: Numpy, Pandas, Scipy, Matplotlib, Seaborn, Flask, FastAPI, OpenCV, Mediapipe, Scikit-Learn, Tensorflow, PyTorch, Natural Language Toolkit (NLTK), Spacy, Dialogflow, Transformers, LangChain, LangGraph, FAISS, Neo4j, Stable-Baselines3, Gymnasium, CrewAI, PocketTTS, Whisper

Developer Tools: GitHub, Hugging Face, Jupyter Notebook, Google Colab, Kaggle, VS Code, Anaconda, Postman, Docker, Kubernetes

Projects

Indic Voice Assistant | *Python, Whisper, IndicTrans2, Docker*

April 2026

- Built an end-to-end multilingual voice pipeline processing Tamil, Hindi, Telugu, Bengali, Tanglish, and Hinglish speech into English audio responses using Whisper ASR, IndicTrans2 and Edge TTS.
- Designed and deployed a FastAPI REST service, Docker support, and a browser-based UI featuring live microphone recording, drag-and-drop upload, and per-stage latency breakdowns.
- Implemented an LRU translation cache reducing repeat IndicTrans2 inference time from 7s to under 10ms, and integrated audio preprocessing (noise reduction, silence removal, normalisation) via librosa and noisereduce.
- Engineered a custom language detection module capable of identifying Unicode scripts and code-mixed dialects (Tanglish/Hinglish) at the segment level, enabling accurate routing across 7 language variants and evaluated model performance using WER and BLEU metrics.

Universal AI Tutor: A Multi Agent System | *LangChain, LangGraph, FastAPI, PocketTTS*

January 2026

- Architected a hybrid AI learning system integrating LLMs + RAG + TTS + Interactive Visualizations, reducing factual errors by 35% compared to baseline LLM responses.
- Implemented Agent Lightning prompt-optimization pipeline, reducing token usage while increasing worksheet generation accuracy and pedagogical structure quality.
- Developed real-time TTS module (PocketTTS integration) for audio responses in low-resource environments.
- Designed multimodal RAG pipeline supporting structured and unstructured data retrieval, increasing response factual reliability.
- Created automated interactive visualization generator (flowcharts, graphs) that improved conceptual clarity scores in user testing.

Marketing Crew | *CrewAI, CrewAI tools, Google-Generativeai, Pyyaml*

October 2025

- Architected AI-powered marketing automation system using CrewAI and Google Gemini 2.5 Flash, orchestrating 4 specialized agents to automate market research, content creation, and SEO optimization workflows.
- Designed sequential multi-agent collaboration system with role-based task delegation, enabling autonomous execution of 8+ marketing tasks with Pydantic-validated structured outputs.
- Integrated web scraping and search APIs with LLM reasoning to automate content research and competitor analysis, reducing workflow time by 80 percent.
- Built modular, YAML-configured system with automated file management, ensuring scalability and maintainability

Research Work

A Comprehensive Evaluation of Embedding Strategies for COVID-19 Fake News Detection

- Conducted a large-scale comparative study of 110+ embedding techniques including BoW, TF-IDF, Word2Vec, GloVe, FastText, BERT, and RoBERTa for fake news detection.
- Presented at SPELLL 2025 and accepted for publication in Springer CCIS.

XAI-driven Evaluation of Embeddings and LLMs for COVID-19 Fake News Detection

- Extended prior work by integrating Large Language Models (LLMs) and Explainable AI (XAI) into the detection framework.
- Accepted for presentation in NIT Goa CVR Conference and publication in Springer.

Abusive Tamil Text Detection Targeting Women on Social Media

- Conducted a comparative study of multilingual and regional NLP models including mBERT, XLM-R, IndicBERT, MuRIL, and L3Cube TamilBERT.
- Ranked 5th in the shared task at ACL DravidianLangTech Workshop, demonstrating strong performance among international research submissions.

COMMUNITY AND LEADERSHIP

Sublead, AI and Data Science Club

October 2024 – October 2025

Indian Institute of Information Technology Kottayam

- Mentored students on various AI, ML, NLP and LLM topics, fostering collaboration and innovation.
- Conducted a hands-on session and guided students in building a project using **OpenCV and Mediapipe**, showcasing advanced computer vision techniques.
- Ideated and organized "*Last Bot Standing*", a national chatbot competition with 150 teams, where participants built puzzle-based secret-revealing bots; led the event end-to-end with successful nationwide participation.